



System Acceptance Review
University Rover Challenge 2026



MONGOL TORI

BRAC University
BRACU Mongol-Tori

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1. Team Overview

1.a. Team Structure: The team BRACU Mongol-Tori consists of 65 undergraduate students from the School of Computer Science, Electrical Engineering, Business, Mathematics, and Natural Science departments of BRAC University, Bangladesh. The team is structured into seven technical sub-teams, under a defined leadership hierarchy of advisor, team lead, and co-lead, collectively responsible for the design, development, and integration of the rover core subsystems. Apart from that, two non-technical sub-teams, Outreach and Finance, are dedicated to influencing school students in robotics and STEM education and securing financial support via strategic collaborations with industries.

1.b. Team Recruitment and Communication: The team follows a structured recruitment process, divided into three phases, comprising screening, self-explanatory task-based evaluation on subsystems, and technical interviews. Selected candidates are trained on lab safety protocols and receive mentorship from senior members to ensure effective knowledge transfer.

2. System Overview and Architecture:

2.a. Mechanical

2.a.i. Chassis and Avionics Enclosure: Taurus, the next generation of the rover is built, as shown in the *Figure 1* from the prior experience of tipping off while on Autonomous Mission (AM), the team opted for a better strength-to-weight ratio by implementing a four-wheel rocker-bogie suspension mechanism with carbon fiber hollow tubes and in-house aluminum extrusion, effectively lowering the overall center of gravity and providing more stability on steep slopes and rocky terrain. Taurus is more compact and lighter than its predecessor, the chassis has been designed to quickly be assembled and disassembled according to the mission requirement. Moreover, the architecture enables future integration of omni-wheel modules for enhanced mobility, while a truss-based enclosure has been built to enhance structural rigidity and to endure the extreme thermal environment of the Mars Desert Research Station (MDRS) while providing modular interfaces for future expansion.

2.a.ii Manipulator: The team has implemented a 7-DOF manipulator in addition to a linear rail to assist the driver in completing precision tasks, especially in USB positioning of Equipment Servicing Mission (ESM). The newly added end effector with myActuator X4-36 motor provides high torque and reliability in precise pitch control in flip switching and lifting a 5 kg payload. Utilizing them, a clip-on 2-DOF autonomous keyboard-typing manipulator has also been developed. Following the prior experience, the team has redefined the adaptive gripper; this will be replaced by an excavator claw during the Science Mission (SM).

2.b. Controls & Software: The control system is divided into two levels: the High-Level Control System (HLCS) and the Low-Level Control System (LLCS). The HLCS decides how Taurus should locomote based on planning, reasoning, and perception-based decision-making. HLCS sends control commands, which include wheel movement and arm movement based on different modes. Based on that, LLCS controls the rover. Based on the similar architecture of ESP32, and keeping in mind the industrial practice, the team has engineered CAN enabled controller, within a small form factor named Kachhim, to control all the actuators. The Inverse Kinematics (IK) of the system has been reengineered by using AS5600 magnetic encoders instead of potentiometers, for more precise angular feedback. This new integration has significantly reduced the computational load as separate CAN-enabled modules read the data and perform PID locally and outputs back to the main controller as shown in the *Figure 2*. The IK system operates in three modes: Full Manual, Full IK, and IK Assisted.

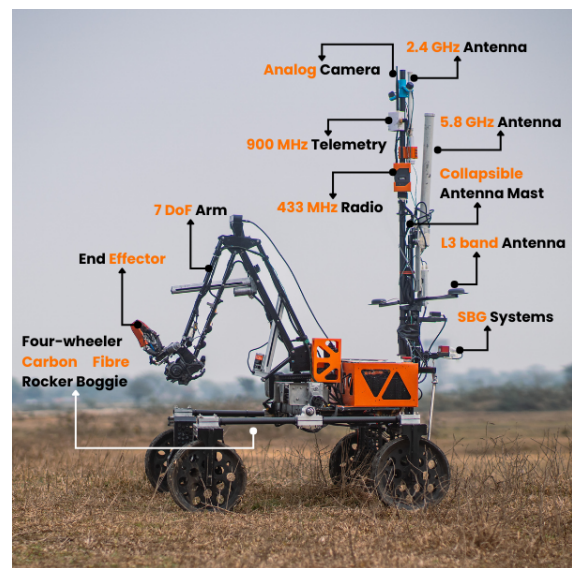


Figure 1: Annotated rover chassis overview.

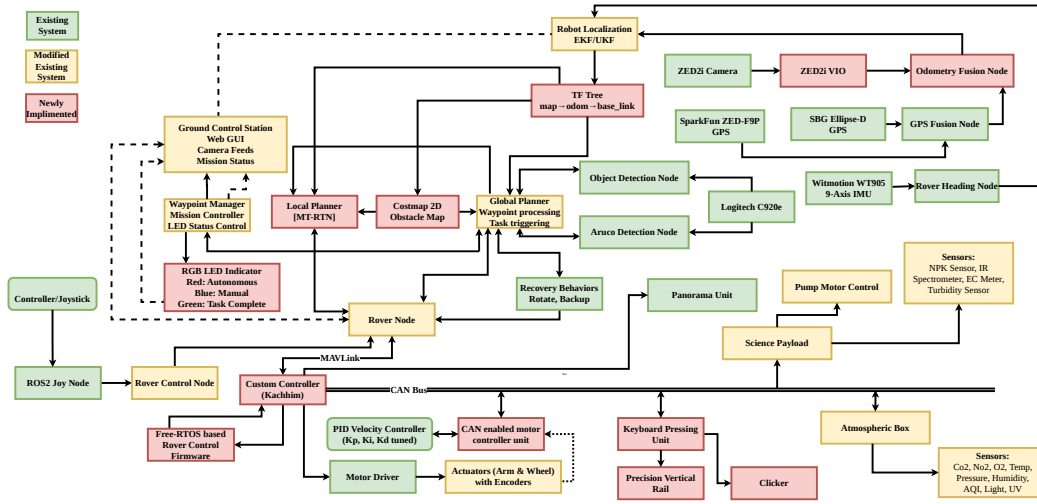


Figure 2: Software system block diagram.

Once the LLCS was tested and validated, mission-specific HLCS was implemented. Like previous iterations, Taurus uses ROS2 Humble with Python for the HLCS. It uses multiple ROS2 packages to maintain control over the rover. They can be categorized into two parts: Mission Specific Packages (MSP) and Common Control Packages (CCP). The MSP decides the rover's actions, and the CCP sends movement commands to the LLCS through telemetry or serial. The GUI is reused from the previous system but has been redesigned significantly, with added system insights, control options, real-time 3D arm visualization, integrated camera feeds, direct IK command input, kill command support, and control mode switching. Each subsystem was tested separately before integration. All subsystems are now integrated, and field testing is ongoing. Full system testing was conducted with all control modes under load conditions to identify faults. Subsystem and system-level tests were performed with the complete rover to detect and resolve errors, bugs, and integration issues.

2.c. Electronics: Adopting the previous year's hot-swappable 18V 9Ah lithium-ion batteries, a critical design choice has been made to eliminate voltage sag issues by introducing multiple isolated voltage rails. It ensures that high draw transients do not induce electrical interference with RF-sensitive components while powering 12V and 24V for communication, 19.5V for Nvidia Jetson, and 5V for logic level. For powering wheels and arm motors, two 60A MDDS60 Bi-Directional DC Motor Driver and four dual channel 10A Cytron MDDS10 are used respectively. Both control and hardware stack are independently powered to prevent unwanted scenarios during power outage. Furthermore, the system incorporates a hardware level reverse polarity protection circuit and a physical kill switch, providing an immediate fail safe against electrical faults or unpredictable autonomous states. The overview of the electronics stack has been depicted in the Figure 3.

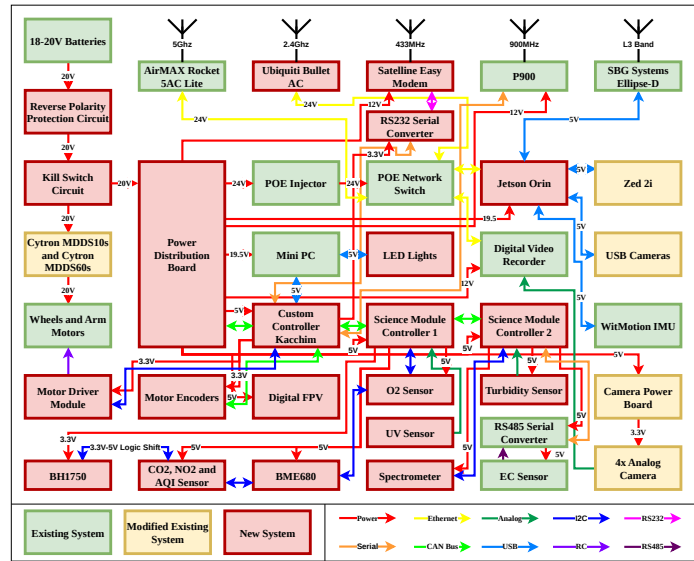


Figure 3: Electronics system block diagram.

2.d. Network and Vision: From the previous years' failure with 5GHz radios for vision during DM, the team has engineered and introduced frequency diversity among radios that will automatically load balance itself and shift to 2.4GHz based on the RSSI, primarily using the 5GHz Ubiquiti airMAX AC Lite router along with a 13dBi airMAX omnidirectional (AMO-5G13) antenna, 2.4GHz Ubiquiti AC Bullet on rover side and 19ddbi (AM-SG120) on base station. The team has also implemented 70cm band with 433MHz SATELLINE

EASy and introduced frequency diversity for mission control as well, along with 900MHz Microhard P900 telemetries have been implemented and tested with 12dBi antennas, making reliable mission control for up to 3.3km at Non-Line of Sight (NLoS) scenarios. These radios signals were simulated, and custom directional and omnidirectional antennas have been built based on the co-ordinates of MDRS ground with the collaboration with CompleTech. The whole system is powered with ProLink Interactive UPS, providing 4 hours backup and real-time power consumption information during mission. This setup has been tested by placing the antennas at 3m height and 25m away from C2 station. Furthermore, 8 analog cameras, along with an in-house digital FPV has been engineered as a failsafe, utilizing SSC338Q camera with OpenIPC firmware, integrated with RTL8812eu Network Interface Card (NIC). “wfb-ng”, a open-source IEEE 802.11-based data-link protocol is used to transmit arbitrary frames without association or acknowledgments, reducing packet drops from protocol timeouts.

2.e. Autonomy: Taurus’s autonomy stack utilizes Differential GNSS with RTK for precise localization, integrated with a 110-degree field-of-view ZED 2i stereo camera for depth perception. Object tracking combines OpenCV ArUco markers with three custom-trained YOLO11n models for hammers and bottle detection. The MT-RTN (Mongol-Tori Rough Terrain Navigation) planner follows a hierarchical state machine architecture, where a global GNSS-based planner manages waypoint navigation and a local planner generates a real-time 3D point cloud costmap for obstacle avoidance. Target acquisition is performed through incremental rotational scanning followed by a square search pattern. An LED indicator on both the rover and base station provides mission-state feedback. The entire system is validated in a custom Unity-based simulator using a digital twin and physics-based environment prior to deployment.

3. Approach to Mission Plan

3.a. Delivery Mission: Taurus has been designed and engineered to specifically traverse precisely through narrow marked path and challenging terrain of DM. Based on experience of previous year, Taurus has been tested in temperatures up to 55°C, and the newly adopted truss design helps keep internal components cool. This also enables attaching a cache box during mission, so that rover can efficiently collect the samples and could deliver them to astronauts. The end effector of Taurus can pick up any tools, rocks, and lid of any heavy object of more than 5 kg in mass. It can also open the hinged lids of the toolbox. The adaptive grip feature of the claw also enables it to hold any object that is greater than 7.5 cm in length. The 12-inch-diameter slotted wheels can easily traverse through sandy terrain and rocky and has been tested through several vertical drops. To achieve better Non-Line of Sight operation, Taurus primarily uses 900MHz and can switch to 433MHz for the mission control. Taurus is equipped with 8 cameras that are transmitted via 5GHz and 2.4GHz, a companion drone has been used to relay the rover’s analog cameras to the control station, along with a retractable antenna mount, which will be lifted up to maintain Line of Sight as much as possible during the mission. The drone also scouts ahead to assist Taurus and read signs. The GUI navigation page assists Taurus to traverse shortest path during the mission. An antenna tracker has also been used to position the directional antennas based on the co-ordinates of the Taurus.

3.b. Equipment Servicing Mission: The manipulator of the Taurus has been designed to perform various maintenance tasks on the rocket lander in a precise way. This year, Taurus features dual end-effector system with independent pitch and roll control, enabling seamless switching between teleoperation and autonomous keyboard-typing tasks. To ensure stability and precision in teleoperating USB-C, linear rail has been implemented, with the added base rotation Taurus can flip switch, carry small hand tools, open latched doors, insert test tubes in the cache box, and connect hose, on its own axis. For the autonomous keyboard typing, Taurus switches to the precision clicker equipped with a vertical axis. The clicker is aligned to the target key through a closed feedback loop using the base linear rail and the vertical rail. Using a camera attached to the clicker, it detects visible keys and decides the approximate position of the target key. Once the target key is in view, it is aligned precisely and a click is performed with the help of a servo motor and the system moves onto the next key, completing the code one character at a time. Multiple cameras has been places in the arm along with a laser crosshair for precise operations.

3.c. Autonomous Navigation: For the AM, Taurus autonomously navigates to GPS-designated waypoints and performs object search tasks under MT-RTN (Mongol-Tori Rough Terrain Navigation), which integrates a GNSS-based global planner, a vision-based local planner, and a task-specific detection system. The global planner follows waypoints from a YAML mission file, continuously computing distance and bearing using the SBG Ellipse-D GNSS and Witmotion WT905 IMU. The rover drives via heading control until reaching a

goal tolerance. In parallel, a ZED 2i stereo system generates a slope-based costmap. If an obstacle lies along the current bearing, control shifts to the local planner, which sweeps headings from -60° to $+60^\circ$, selects the safest direction biased toward the goal, rotates, and advances while monitoring clearance. Once clear, control returns to the global planner. A GPS accuracy monitor pauses motion if fix quality degrades. At each waypoint, the mission coordinator triggers the assigned task. ArUco markers are detected using OpenCV's ArucoDetector, while mallet and bottle tasks use custom YOLO models. If no target is visible, the rover performs a 360° rotational search. Upon detection, it estimates distance, centers the target, and approaches to 1.5 m before stopping. Mission progress is indicated via the onboard LED matrix, and a custom GUI is used to define GNSS waypoints, mark search areas, and visualize live tracking throughout operation.

4. Testing & Operation:

In preparation for the 2026 University Rover Challenge (URC), Team Mongol-Tori undertook rigorous training and testing to establish team-wide proficiency across all aspects of competitive rover design and operation. The mechanical subteam completely redesigned the 2025 rover, Hypersonic, utilizing SOLIDWORKS to engineer the new primary rover, Taurus, retaining only the original chassis. To ensure structural integrity and optimize mass, all mechanical designs underwent comprehensive finite element analysis (FEA) and topological optimization using Ansys before manufacturing. Concurrently, the controls and electronics subteams developed a custom, in-house controller board based on the ESP32 platform, incorporating CAN bus submodules to integrate all sensors and actuators into a unified communication network.

To acclimate new members to the existing software stack, the autonomous and controls subteam hosted a series of technical workshops. Furthermore, recruits were integrated into their respective subteams under the mentorship of senior leads and provided with a fully functional secondary clone of Taurus. This clone facilitated simultaneous testing and development, allowing junior members to safely execute programs and cultivate competition-ready skills without halting parallel development on the primary rover. System testing was conducted in a phased approach: active construction grounds adjacent to the Brac University campus were utilized for short-range, rapid hardware validation, while nearby rocky, uneven terrains and suburban fields served as proving grounds for long-range teleoperation and URC mock simulations.

These concentrated development cycles allowed the team to advance the rover's NASA Technology Readiness Level (TRL) from TRL-1 to TRL-5 within a nine-month timeframe. Moving forward, comprehensive system testing and operator training will continue. Additionally, detailed operating procedures are being refined for each URC mission to define clear success criteria and establish robust fallback modes in the event of unexpected technical challenges.

5. Project Management:

The project follows a structured formulation, development, and implementation life cycle model. The two primary phases being formulation and implementation. These phases are dependent on the different sub teams; each dedicated to specific rover subsystems, as shown in *Figure 4*. Each sub teams functions independently, merging during the system integration phase, using the previous variant as a baseline metric. Each technical sub team is subjected to comprehensive performance assessments to ensure that it is prepared for the URC grounds. A detailed budget breakdown is presented in *Figure 5*, showing the income and expenses of the team as of February 27, 2026.

Science Mission

The atmospheric module is redesigned and constructed to maximize accuracy and efficiency. Different parameters are correlated with onboard laboratory experiments to strengthen the proof of site habitability. For temperature, humidity, atmospheric pressure, and air quality index sensor (BME 680), Electrochemical O₂ sensor, SCD41 Infrared Carbon Dioxide sensor, VEML7700 Ambient Light Sensor, UV Index Sensor, PM2.5 Air Quality Sensor, Fermion: MEMS Methane CH₄ Gas and Nitrogen Dioxide NO₂ Detection Sensor are calibrated to match our demand, depending on the site. A high-resolution camera mounted to a servo motor is used to capture images from multiple directions and are stitched together to produce a panorama image for stratigraphic profiling.

Subsurface soil analysis depends on geological features observed such as humid soil samples or cracks on the surface, which indicate an easy drilling method. Drill site condition is analyzed via RS485 MODBUS-RTU soil NPK (Nitrogen, Phosphorus, Potassium) to measure calibrated sensor for pre-analysis. Interlocking soil collecting claws, NPK sensors and auger are added to the science arm at 90-degree rotation in a single axis to ensure fast data interpretation per site. Mathematical proof ensures the drilling of soil more than 15-centimeter, one-sided valve induced load cell added cache collecting cuvette is 3D printed for no contamination policy.

Life detection experiment is divided into three divisions: Organic matter test by Hydrogen peroxide for understanding the presence of soil organic matter after the presence of oxidation reaction which helps to determine the presence of extinct life. After harsh temperature conditions, oxalate organic matter presence and the possibility of life organisms to thrive is the getaway.^[1]

Electric conductivity (EC) test of soil indicates the presence of extant life present in any viable sample as it determines the dielectric properties of soil.^[2] Key gateway from this experiment shows the frequency response of applied sinusoidal voltage to understand the properties of granular materials present.^[3] Voltage difference and differential graph obtained is very crucial for understanding the extant life formation capability of the test site.^[4]

Phosphate test by Olsen method^[5], as shown in *Figure 6* detects the possibility of life formation and strengthens it by revealing igneous and aqueous condition of site and opening prebiotic chemistry of DNA, RNA and ATP synthesis. Igneous processes enhance phosphorus in Martian magmas to varying degrees, and igneous phosphates (IP) are frequent in late-stage crystalline assemblages.^[6] IP can potentially provide crucial information about Mars' igneous history. The presence of phosphorus (P) in Martian rocks may have a role in the development of iron oxides like hematite (α -Fe₂O₃) and maghemite (γ -Fe₂O₃), which are found in Martian soil. This idea is confirmed by laboratory tests that show that thermal transition of P-doped lepidocrocites and magnetites produces hematite and maghemite with magnetic and spectral properties like those found in Martian soils.^[7]

Further identification of carbohydrates and nitrogen is done by UV-C ray analysis. Proteins primarily absorb UV light due to the presence of tryptophan and tyrosine with molar absorption coefficient at 280 nm. UV absorbance of a protein depends on the concentration and arrangement of these amino acids. The simplest way to quantify proteins is by measuring UV absorbance using a UV-C sensor, typically between 260 and 280 nm, making it ideal for pure protein detection.^[8]



Figure 6: Annotated science module overview.

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